

Hikmalayer Core

Local Docker Testnet Deployment & Test Suite Report

Date	14 July 2026
Repository	bestowerlabs/hikmalayer-core
Environment	Windows 11 + WSL2 (Ubuntu 26.04) + Docker Desktop (WSL2 backend)
Hardware	8 GB RAM host; WSL2 capped to 5 GB via .wslconfig

Summary

The full 9-service Docker Compose testnet (bootnode, 4 validators, RPC node, Prometheus, Grafana, JSON exporter) was built and deployed locally and confirmed healthy end-to-end. The Rust test suite was run in parallel and passed in full.

Deployment	9 / 9 containers Up
Validators registered on-chain	3
Chain height reached	4
Unit / integration tests	70 / 70 passed
Test failures	0

1. Local Deployment Result

Deployed via `./ops/start_testnet.sh` with default P2P_TOKEN / ADMIN_TOKEN (local-testnet / local-admin).

Containers (all reached a healthy Up state):

Container	Service	Port
hikmalayer-bootnode	bootnode	3000
hikmalayer-validator1	validator1	3001
hikmalayer-validator2	validator2	3002
hikmalayer-validator3	validator3	3003
hikmalayer-validator4	validator4	3004
hikmalayer-rpc	rpc	3010
hikmalayer-json-exporter	json_exporter	7979

hikmalayer-prometheus	prometheus	9090
hikmalayer-grafana	grafana	3005

On-chain validator set:

```
hkm016bfe32723b4ceab2584aecceca763afc2c2e69    stake: 100
hkm50929b74c1a04954b78b4b6035e97a5e078a5a0f    stake: 1000
hkm663c1b408cf896dd00e20c84b2e734b5d8da0893    stake: 100
```

Blockchain state:

```
height: 4 | total_supply: 1,000,020 | burned: 0 | staked: 1,200 | validators: 3 |
accounts: 4
state_root: 95e26ee5ca32eb7388960250c7a9826ef0be9863ccab5d6c0ea3d19001c40890
```

Issue found & fixed during deployment:

- The repository's Dockerfile entrypoint (CMD ["cargo", "run", "--release"]) failed at container startup with error: 'cargo run' could not determine which binary to run, once the hikma-wallet binary was added alongside the main hikmalayer binary. Fixed by pinning the entrypoint to CMD ["cargo", "run", "--release", "--bin", "hikmalayer"].

Recommend committing this fix so future clean builds/CI don't hit the same failure.

Confirms end-to-end local deployment: build, container orchestration, faucet funding, on-chain staking, mining, and state root computation all functioning correctly.

2. Test Suite Result

Ran the full Rust test suite with `cargo test -j 2` (parallel job count reduced to fit an 8 GB RAM host; default parallelism triggered `Cannot allocate memory (os error 12)` during compilation).

Total tests run	70
Passed	70
Failed	0
Ignored	0
Duration	1.01s

Coverage highlights:

- Blockchain core: block creation, tampering detection, fork-choice, difficulty retargeting
- Consensus: Proof-of-Work, Proof-of-Stake selection, VRF randomness beacon
- Security-relevant: P2P token rejection/replay protection, forged/unsigned block rejection, admin/treasury key requirements on faucet, credential lifecycle
- Networking: peer scoring, banning on repeated misbehavior, envelope replay protection

A separate hikma-wallet unit-test binary also ran as part of the same invocation; its pass/fail summary should be confirmed independently before treating the full suite as fully green.

3. Environment Notes

- WSL2 required explicit integration enablement for the Ubuntu distro under Docker Desktop Settings → Resources → WSL Integration.
- The Docker socket required the local user to be added to the docker group (`sudo usermod -aG docker $USER`), followed by a full WSL restart (`wsl --shutdown`).
- Rust/Cargo needed to be installed separately inside WSL (a Windows-side `rustup` install is not visible to the WSL filesystem/environment).
- Project files were relocated out of a OneDrive-synced folder before building, to avoid file-locking issues during Docker image builds.
- WSL2 memory was capped to 5 GB via a `.wslconfig` file on the Windows host to prevent out-of-memory failures on this 8 GB machine.